

A Lightweight Lattices inspired Bi-cubic Hermite Surface Construction based Learning With Errors Public Key Cryptosystem

Gopal Anantharaman*
KnotTheory.ai Inc.,
Fremont, CA, USA

August 4, 2025

Abstract

A fast public key encryption scheme is proposed using a secondary square matrix with entries from Z_p , p being prime, to sample the two vectors used in the LWE randomly [3] uniformly for use in public and private key constructions. We transform the secondary square matrix to a bi-cubic Hermite Patch, sometimes degenerate patches, giving random slopes and other closed curves for each matrix entry - essentially constructing an open set around each entry(which acts as lattice points). We then use the complexity inherent in the geometry of gerrymandering [1], [2] of a 2-D matrix to generate gerrymandered units of the transformed matrix. From those pieces, we construct two uniform random vectors s_{priv} and s_{pub} using Vitters' Algorithm [4] for use in LWE-based single-party key constructions. We also show that in Multi-party communication, different transformations of the secondary matrix can be used to sample the s vectors for multi-party communication with negligible attack concerns using the same secondary matrix. The secondary matrix, along with a Bi-cubic Hermite patch transformational polynomials, could be issued to trusted entities by a Certifying Authority

1 Introduction

Learning

*E-mail: gopal.anantharaman@knottheory.ai; Corresponding author

2 LWE Scheme

3 Definitions

4 Encryption/Decryption

5 Cryptanalysis

6 Strength Analysis

7 Extensions and Complexity

An example of a gerry

1023	23	67	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
Q	W	E	R	T	Y	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	A	B	c	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R
A	B	K	H	E	F	C	S	O	S	R	R	R	A	B	K	H	E	F	C	S	O	S	R	R	R

Thus, Stencils provide enough obfuscation atop fast crypto-schemes.

8 Conclusion

Modern camera-based AI agent communication with vision helps process data asymmetrically using Cameras, Network protocols, and Human Collaboration. In such a Hybrid environment, seamless, secure transactions across domains are becoming a necessity rather than a theoretical interest while retaining existing Encryption schemes. As shown, The MHKC or other fast encryption schemes could be retrofitted with simple obfuscations to achieve very reliable security mechanisms for Machine-to-Machine or Human-to-Machine communications.

References

- [1] Frunobulax (<https://mathoverflow.net/users/57621/frunobulax>). *Is this kind of "Gerrymandering" NP-complete?* MathOverflow. URL:<https://mathoverflow.net/q/324524> (version: 2020-07-17). eprint: <https://mathoverflow.net/q/324524>. URL: <https://mathoverflow.net/q/324524>.
- [2] Rectilinear Steiner Tree. *Rectilinear Steiner Tree*. [Online; Accessed 2 Jul. 2024]. 2024. URL: http://en.wikipedia.org/wiki/Rectilinear_Steiner_tree.
- [3] Oded Regev. "On lattices, learning with errors, random linear codes, and cryptography". In: *J. ACM* 56.6 (Sept. 2009). ISSN: 0004-5411. DOI: 10.1145/1568318.1568324. URL: <https://doi.org/10.1145/1568318.1568324>.
- [4] Jeffrey Scott Vitter. "Faster methods for random sampling". In: *Commun. ACM* 27.7 (July 1984), pp. 703–718. ISSN: 0001-0782. DOI: 10.1145/358105.893. URL: <https://doi.org/10.1145/358105.893>.